

Student ID 1 & 2: (Do this in pairs)

A. Recall that in Union by Rank, the root with smaller rank is attached under the root with larger rank. The rank only increases when two equal-rank trees are merged. **Prove that Union by rank results in $O(\log n)$ cost per Find in Union-Find Data Structure.**

Hint: Show that a tree of rank r must contain at least 2^r nodes. Use this to bound the maximum possible height.

B. Design a decimal counter and perform amortized analysis of number of digits that need to be changed per increment (in class we did this for binary counter.) The counter is in base 10, with digits 0–9. When a digit becomes 9, it resets to 0 and generates a carry.

Hint: Count how many times each digit position changes over n increments. First compute the total number of digit changes across all n increments, and then divide by n to obtain the amortized cost per increment.